International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Mid Exam, Spring-2025

Course Code: CSE 3631 **Course Title**: Operating System

Time: 1 hours 30 minutes **Full Marks**: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1. a) How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security)? C01 03

b) Explain the layered approach to operating system design. What are the advantages and disadvantages of using this approach? CO1 04

c) What are some advantages of peer-to-peer systems over client–server systems? C01 03

2. a) In a mobile operating system (like Android), background and foreground tasks are managed using a hybrid scheduling algorithm. High-priority tasks (e.g., UI rendering, audio playback) are given preemptive priority over background services (e.g., syncing data, notifications). However, to ensure fairness among tasks of the same priority (e.g., multiple background apps), Round Robin is used. CO2 07

The system uses: Three priority levels: High (UI/audio), Medium (background apps), Low (background daemons) and RR time quantum: 10 ms for each level. Consider the following scenario :

Task	**Arrival**	**Burst Time**	**Priority**	**Description**
UI Thread	0 ms	30 ms	High	Touch input + UI rendering
Sync App	5 ms	20 ms	Medium	Google Drive background sync
Music App	7 ms	25 ms	High	Audio playback
Cleanup Daemon	10 ms	15 ms	Low	Periodic log cleanup
Chat Notification	12 ms	10 ms	Medium	Push notification background task

Now, answer the following questions :

i) Simulate the execution(Gantt Chart) of these tasks for using hybrid Priority + RR scheduling (RR quantum = 10 ms).

ii) Calculate the average waiting time.

iii) In the scenario above, identify potential starvation risks for the low-priority background daemon. Explain how the hybrid RR + Priority approach contributes to or mitigates starvation.

b) Describe the actions taken by a kernel to context-switch between processes. C01 03

3. a) Explain the concept of Process Control Block data structure, that is maintained by the operating system to keep track of a particular process. CO2 02

b) A cloud computing platform is processing multiple tasks with varying burst times. The server uses Shortest Remaining Time First (SRTF) scheduling to ensure that tasks with the shortest remaining burst time are processed first. The tasks are submitted at different times, and the platform needs to decide how to schedule them. CO2 05

Task	Arrival Time (ms)	Burst Time (ms)
T1	0	12
T2	2	8
T3	4	6
T4	5	10

Simulate the execution of these tasks using SRTF scheduling.

- Draw the Gantt chart.
- Calculate the waiting time and turnaround time for each task.
- Analyze how preemption works in SRTF when new tasks arrive while others are still executing. Which task experiences the most preemption, and why?

c) A drone navigation system uses EDF to manage tasks related to sensor updates, trajectory computation, and battery management. C02 03

- **T1 (Sensor update):** Execution = 1 ms, Relative Deadline = 5 ms
- **T2 (Trajectory computation):** Execution = 3 ms, Relative Deadline = 10 ms
- **T3 (Battery check):** Execution = 1 ms, Relative Deadline = 8 ms

i)Simulate the execution of these tasks over a 20ms interval using EDF. Analyze how deadlines are prioritized and whether any tasks miss their deadlines.

ii)Compare the EDF schedule with a Rate Monotonic schedule for the same task set. Which scheduling policy better suits this drone application, and why?

"You who believe, seek help through steadfastness and prayer, for God is with the steadfast." [Quran 2:153]